

VAPT INTERNSHIP

WEEK 3 Anonymity, VPN & OSINT

Sulav Adhikari

1. SECTION A - Multiple Choice Questions

VAPT Internship — Nepal | Week 3

Quiz: Anonymity, VPN & OSINT

sulavadh124@gmail.com [Switch accounts](#)

Not shared [Draft saved](#)

* Indicates required question

SECTION A – MCQs (20 Marks)

1. What does OPSEC stand for? *

- ☒ a) Operational Security
- ☐ b) Open Security
- ☐ c) Online Protection System
- ☐ d) Offensive Security

2. Which is the first step in OPSEC? *

- ☐ a) Apply Countermeasures
- ☒ b) Identify Critical Information
- ☐ c) Assess Risk
- ☐ d) Analyse Threats

3. Which of the following is considered critical information? *

- ☐ a) Public website data
- ☒ b) IP addresses and tools used
- ☐ c) Company logo
- ☐ d) Social media posts

4. Who can potentially monitor your activity? *

- ☐ a) ISP
- ☐ b) System Administrator
- ☐ c) Law Enforcement
- ☒ d) All of the above

5. What is a common OPSEC vulnerability? *

- ☐ a) Strong password
- ☐ b) Encrypted traffic
- ☒ c) DNS leaks
- ☐ d) Firewall rules

VAPT Internship — Nepal | Weebly

docs.google.com/forms/d/e/1FAIpQLSd2ChwiOumZEtetL6Eq-u00oakYDeoPloemenFSOZsyft_GNA/formResponse?pli=1

LMarena into+litrev_prototyp... Nepali New Year Pr...

☐ d) Firewall rules

6. OPSEC failures usually happen due to: *

☐ a) Advanced hacking

☐ b) Expensive tools

☒ c) Human mistakes

☐ d) Network speed

7. What does a VPN primarily do? *

☐ a) Deletes logs

☒ b) Encrypts traffic

☐ c) Blocks malware

☐ d) Speeds up internet

8. A commercial VPN provides *

☐ a) Full anonymity

☒ b) One-hop privacy

☐ c) No encryption

☐ d) Direct connection

VAPT Internship — Nepal | Weebly

docs.google.com/forms/d/e/1FAIpQLSd2ChwiOumZEtetL6Eq-u00oakYDeoPloemenFSOZsyft_GNA/formResponse?pli=1

LMarena into+litrev_prototyp... Nepali New Year Pr...

☐ d) Direct connection

9. Which VPN type connects two networks? *

☐ a) Split tunnel

☐ b) Commercial VPN

☒ c) Site-to-site VPN

☐ d) Proxy VPN

10. Split tunneling means: *

☐ a) All traffic goes through VPN

☐ b) No traffic goes through VPN

☒ c) Some traffic uses VPN

☐ d) Only DNS traffic uses VPN

11. Which is faster and modern? *

☐ a) OpenVPN

☒ b) WireGuard

☐ c) IPSec

☐ d) PPTP

VAPT Internship — Nepal | Weebly x +

docs.google.com/forms/d/e/1FAIpQLSd2ChwiOumZEtecL6Eq-u00oakY0eoPloemenFSOZsyft_GNA/formResponse?pli=1

12. In Tor, which node knows your IP? *

- ☐ a) Exit node
- ☐ b) Relay node
- ☒ c) Guard node
- ☐ d) Target server

13. Which node knows the destination? *

- ☐ a) Guard
- ☐ b) Relay
- ☒ c) Exit
- ☐ d) Client

14. Tor provides anonymity through: *

- ☐ a) Encryption only
- ☒ b) Onion routing
- ☐ c) Firewalls
- ☐ d) Antivirus

VAPT Internship — Nepal | Weebly x +

docs.google.com/forms/d/e/1FAIpQLSd2ChwiOumZEtecL6Eq-u00oakY0eoPloemenFSOZsyft_GNA/formResponse?pli=1

15. ProxyChains is used to: *

- ☐ a) Scan faster
- ☐ b) Encrypt files
- ☒ c) Route tools through proxies
- ☐ d) Detect malware

16. Which tool is used for email harvesting? *

- ☐ a) Shodan
- ☐ b) Maltego
- ☒ c) theHarvester
- ☐ d) Nmap

17. What does Shodan index? *

- ☐ a) Websites only
- ☒ b) Internet-connected devices
- ☐ c) Emails
- ☐ d) Passwords

VAPT Internship — Nepal | Week 3

docs.google.com/forms/d/e/1FAIpQLSd2ChwiOumZEtetL6Eq-uooakY0eoPloemenFSOZsyft_GNA/formResponse?pli=1

LMARena into+litrev_prototyp... Nepali New Year Pr...

☐ d) Passwords

18. Which Google dork finds login pages? *

☐ a) site:
☒ b) inurl:login
☐ c) filetype:pdf
☐ d) intext:password

19. OSINT is: *

☐ a) Active scanning
☒ b) Passive information gathering
☐ c) Exploitation
☐ d) Privilege escalation

20. A recon report must include: *

☐ a) Only screenshots
☐ b) Only tool output
☒ c) Structured findings
☐ d) Just IP addresses

2. SECTION B - Short Answer Questions

VAPT Internship — Nepal | Week 3

docs.google.com/forms/d/e/1FAIpQLSd2ChwiOumZEtetL6Eq-uooakY0eoPloemenFSOZsyft_GNA/formResponse?pli=1

LMARena into+litrev_prototyp... Nepali New Year Pr...

VAPT Internship — Nepal | Week 3
Quiz: Anonymity, VPN & OSINT

sulavadh124@gmail.com [Switch accounts](#)

Not shared Draft saved

* Indicates required question

SECTION B – LONG QUESTIONS (20 Marks)

Q1. Explain the 5-step OPSEC process in detail. (5 marks) *

The 5-step OPSEC (Operational Security) process helps protect sensitive information:

- Identify Critical Information**
-Determine what information must be protected (e.g., IP addresses, system details, credentials). This is the most important step because everything else depends on it.
- Analyze Threats**
-Identify potential adversaries (hackers, competitors, insiders) and understand their capabilities and intentions.
- Analyze Vulnerabilities**
-Find weaknesses that could expose critical information, such as poor passwords, misconfigurations, or DNS leaks.
- Assess Risk**
-Evaluate the likelihood and impact of threats exploiting vulnerabilities. This helps prioritize which risks need immediate attention.
- Apply Countermeasures**
-Implement security controls (VPNs, encryption, firewalls, policies, training) to reduce or eliminate risks.

Q2. Explain the working of VPN and its types. (5 marks) *

VPN (Virtual Private Network) creates a secure and encrypted connection over the internet.

Working of VPN:

- When a user connects to a VPN:
- a. The device establishes a secure tunnel to a VPN server.
- b. All internet traffic is encrypted before leaving the device.
- c. The VPN server decrypts the data and forwards it to the destination.
- d. The destination sees the VPN server's IP instead of the user's real IP.
- This ensures privacy, security, and anonymity (to some extent).

Types of VPN:

- a. Remote Access VPN
 - Allows individual users to connect securely to a private network (commonly used by employees working remotely).
- b. Site-to-Site VPN
 - Connects two or more networks (e.g., branch office to headquarters).
- c. Commercial VPN
 - Provided by companies (like NordVPN, ExpressVPN) for general privacy; typically offers one-hop encryption.
- d. Split Tunneling VPN
 - Allows some traffic to go through VPN and some through the regular internet.

Q3. Explain the Tor network architecture. (5 marks) *

Tor (The Onion Router) provides anonymity using layered encryption and multiple nodes.

Architecture Components:

- a. Client (User)
 - The user runs Tor Browser to access the network.
- b. Guard Node (Entry Node)
 - The first node in the circuit.
 - It knows the user's IP but not the destination.
- c. Relay Node (Middle Node)
 - Passes encrypted traffic between nodes.
 - It does not know the source or destination.

- Passes encrypted traffic between nodes.
- It does not know the source or destination.
- d. Exit Node
 - The final node in the circuit.
 - It decrypts the last layer and sends traffic to the destination server.
 - It knows the destination but not the user.

Working (Onion Routing):

- Data is encrypted in multiple layers (like an onion).
- Each node removes one layer of encryption.
- Traffic passes through Guard → Relay → Exit node.
- No single node knows both the source and destination.
- This ensures strong anonymity.

Q4. Explain OSINT tools and techniques. (5 marks) *

OSINT (Open Source Intelligence) is the process of collecting publicly available information for investigation or reconnaissance.

Common OSINT Tools:

- a. theHarvester
 - Used to collect emails, subdomains, and hostnames from public sources.
- b. Shodan
 - Searches internet-connected devices such as servers, cameras, and IoT devices.
- c. Maltego
 - Visual tool used to map relationships between people, domains, and organizations.
- d. Google Dorking
 - Uses advanced search queries (e.g., inurl:login filetype:pdf) to find sensitive information.

OSINT Techniques:

- a. Passive Reconnaissance
 - Gathering information without directly interacting with the target.
- b. Search Engine Analysis
 - Using search engines like Google to find indexed data.
- c. Social Media Analysis
 - Collecting information from platforms like LinkedIn, Facebook, etc.
- d. Metadata Analysis
 - Extracting hidden information from files (e.g., author, location).

3. SECTION C - Practical Submission Tasks

a. Task C1 - OSINT Recon Report

Target Domain: testphp.vulnweb.com

Information Type	Your Findings	Tool / Source Used
Registrar & Registration Date	Domain is registered under Gandi SAS . The domain creation date is 14 June 2010 .	WHOIS

Nameservers (NS records)	NS-105-A.GANDI.NETNS-11-B.GANDI.NETNS-140-C.GANDI.NET	WHOIS
IP Address(es)	The domain resolves to multiple IP addresses: 44.228.249.3, 176.28.50.165, 18.192.172.30	nslookup + theHarvester
Subdomains / URLs found	TheHarvester and analysis revealed web endpoints: testphp.vulnweb.com, login.php, artists.php, listproducts.php	theHarvester
Email addresses found	No email addresses were discovered during passive reconnaissance.	theHarvester
Technologies / Server software	The target web application is likely running on Apache HTTP Server with PHP-based backend technology .	Shodan (attempted)
Any exposed services (Shodan)	No exposed services were found. The target is not indexed in Shodan or does not expose publicly visible services.	Shodan
Interesting Google Dork finding	Google Dorking revealed a login page: “Acunetix VulnWeb Login Portal” using query site:testphp.vulnweb.com inurl:login	Google Dork

- WHOIS Lookup (Domain Information)

Command: whois vulnweb.com

```

kali@kali: ~
$ whois vulnweb.com
Domain Name: VULNWEB.COM
Registry Domain ID: 1602006391_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.gandi.net
Registrar URL: http://www.gandi.net
Updated Date: 2025-11-17T09:34:20Z
Creation Date: 2010-06-14T07:50:29Z
Registry Expiry Date: 2027-06-14T07:50:29Z
Registrar: Gandi SAS
Registrar IANA ID: 81
Registrar Abuse Contact Email: abuse@support.gandi.net
Registrar Abuse Contact Phone: +33.170377661
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Name Server: NS-105-A.GANDI.NET
Name Server: NS-11-B.GANDI.NET
Name Server: NS-140-C.GANDI.NET
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-04-13T06:05:32Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

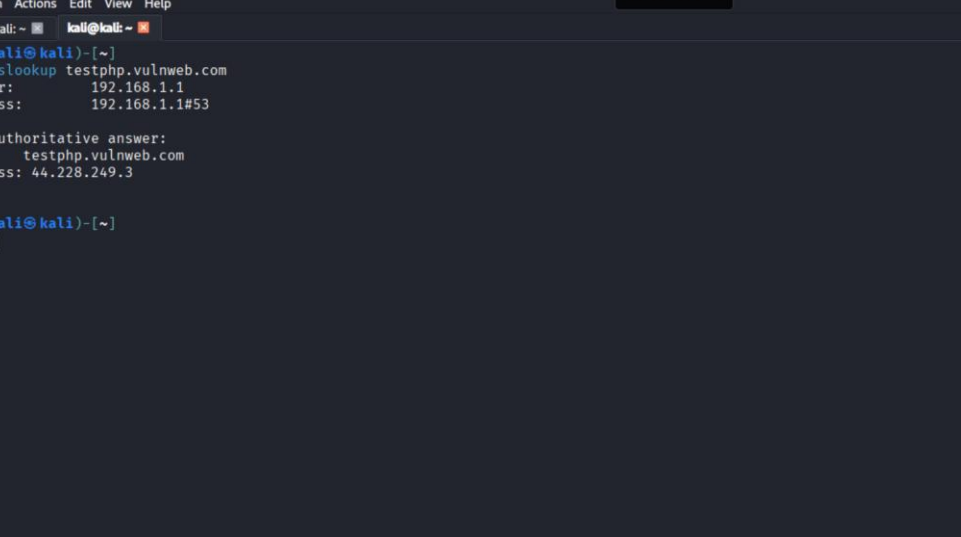
NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or

```

- Command: nslookup testphp.vulnweb.com

Command: nslookup testphp.vulnweb.com



The screenshot shows a Kali Linux terminal window. The title bar at the top includes standard Linux window controls and application-specific icons. The terminal interface has a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. Below the menu bar, the prompt 'kali@kali: ~' is visible, followed by a tab labeled 'kali@kali: ~'. The user has entered the command 'nslookup testphp.vulnweb.com'. The terminal output shows the results of the DNS lookup: 'Server: 192.168.1.1', 'Address: 192.168.1.1#53', and a 'Non-authoritative answer:' section with 'Name: testphp.vulnweb.com' and 'Address: 44.228.249.3'. The prompt '(kali@kali)-[~]' is shown again after the output. A status bar at the bottom right indicates 'CPU usage: 2.0%'.

```
kali@kali: ~  
kali@kali: ~  
(kali@kali)-[~]  
$ nslookup testphp.vulnweb.com  
Server:      192.168.1.1  
Address:     192.168.1.1#53  
  
Non-authoritative answer:  
Name:   testphp.vulnweb.com  
Address: 44.228.249.3  
  
(kali@kali)-[~]  
$
```

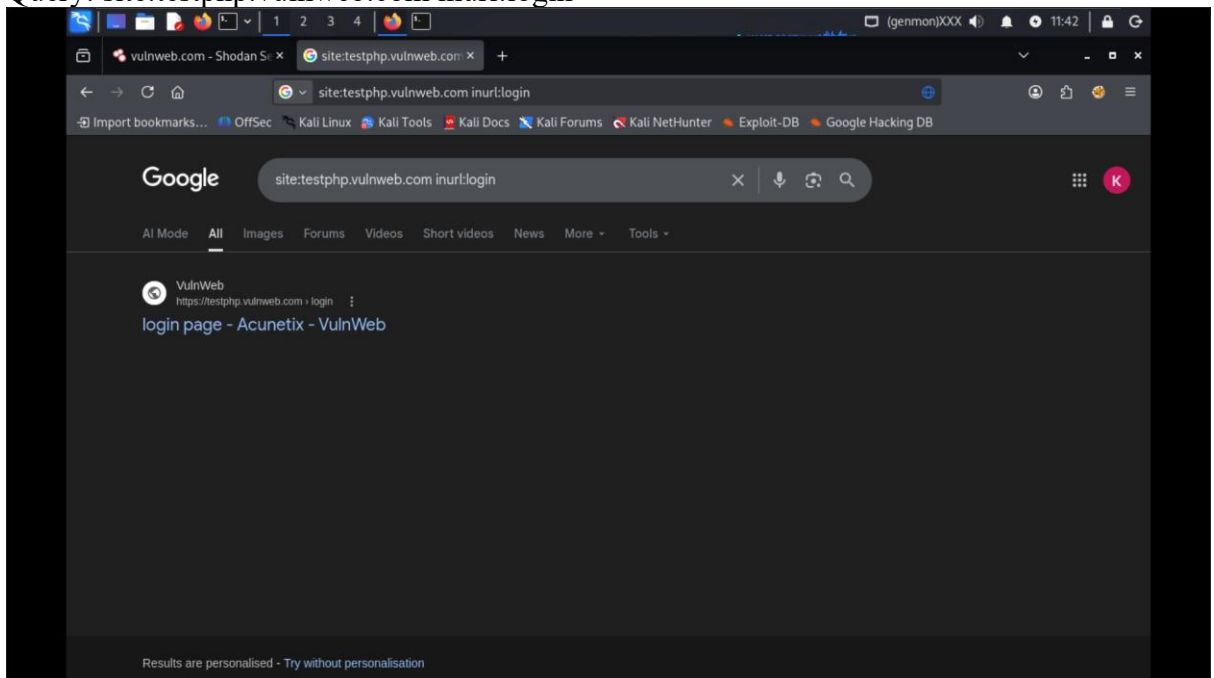
- Command: theHarvester -d vulnweb.com -b crtsh

Command: theHarvester -d vulnweb.com -b crtsh

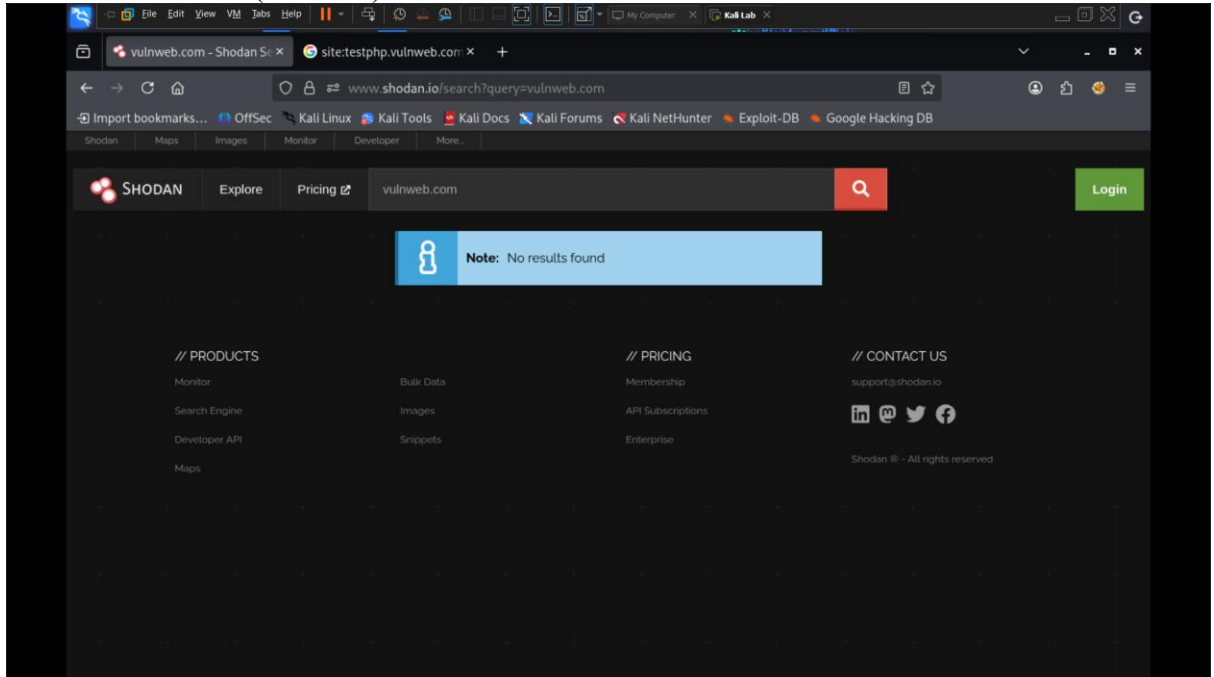
[illegible]

```
kali@kali: ~  
[*] Interesting Urls found: 6  
http://testphp.vulnweb.com/  
http://testphp.vulnweb.com/artists.php?artist=1  
http://testphp.vulnweb.com/listproducts.php?cat=1  
http://testphp.vulnweb.com/login.php  
https://testphp.vulnweb.com/  
https://testphp.vulnweb.com/login.php  
[*] No LinkedIn users found.  
[*] LinkedIn Links found: 0  
[*] IPs found: 4  
176.28.50.165  
18.192.172.30  
44.228.249.3  
[*] No emails found.  
[*] No people found.
```

- Google Dork (Browser Search)
Query: `site:testphp.vulnweb.com inurl:login`

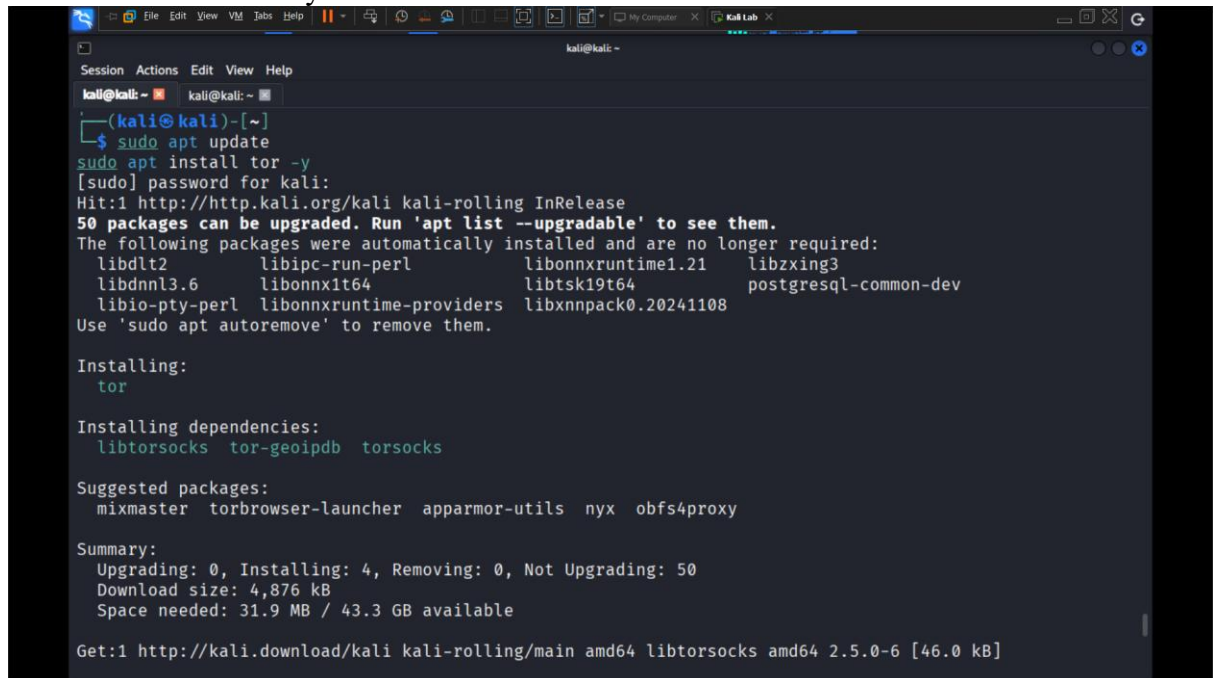


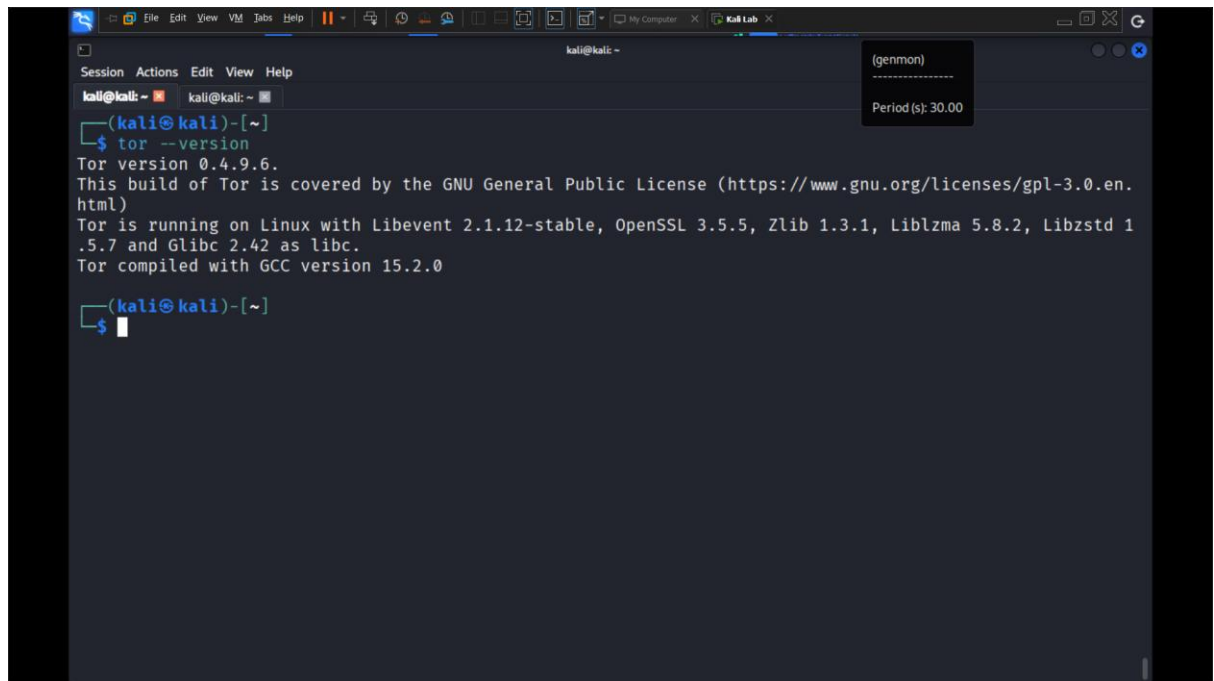
- Shodan Search (Browser)



b. Task C2 - ProxyChains & Tor Setup

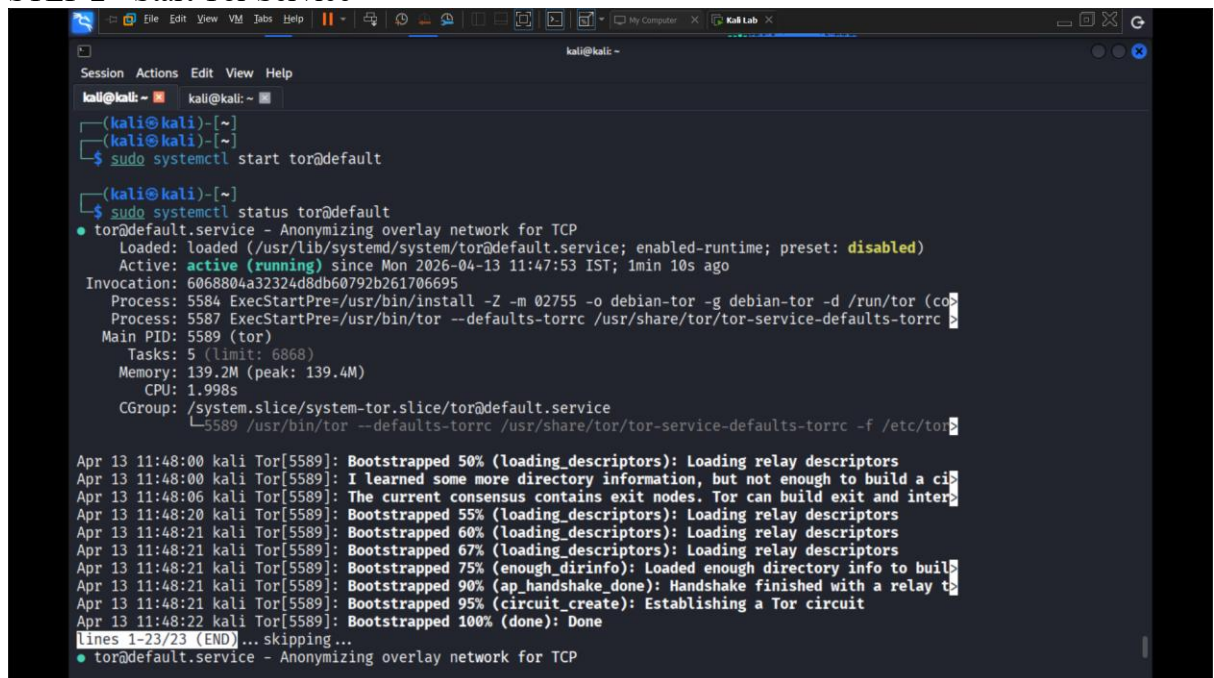
- STEP 1 - Install / Verify Tor





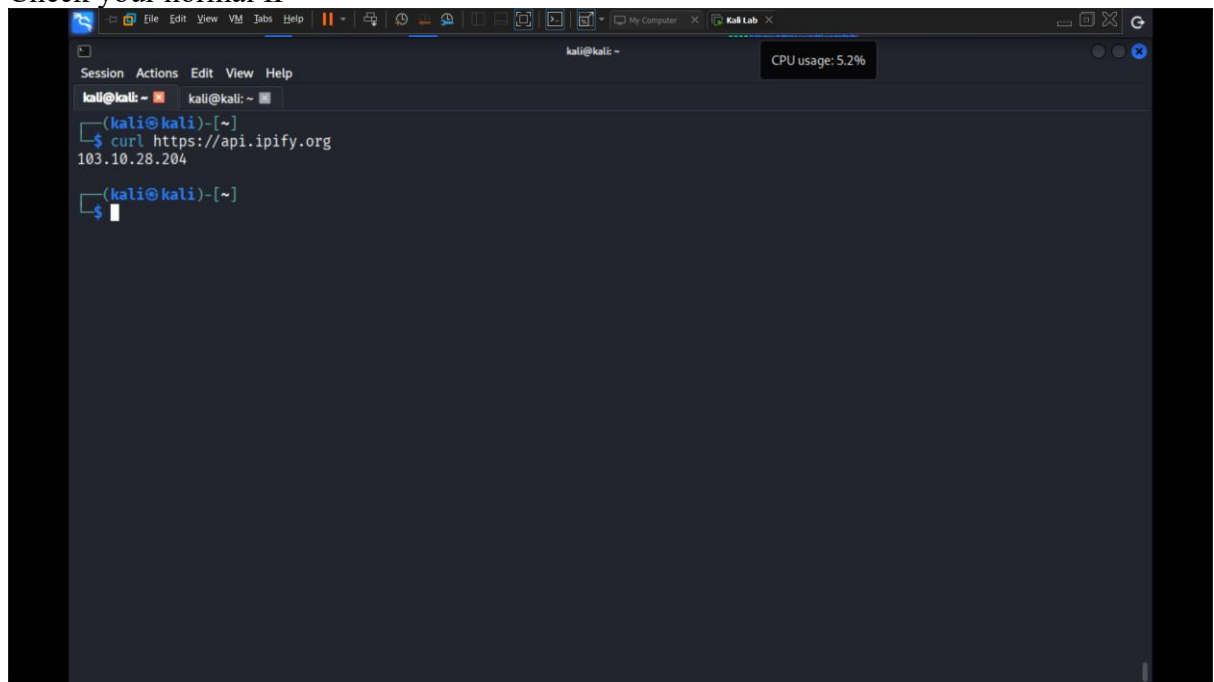
```
kali@kali: ~  
$ tor --version  
Tor version 0.4.9.6.  
This build of Tor is covered by the GNU General Public License (https://www.gnu.org/licenses/gpl-3.0.en.html)  
Tor is running on Linux with Libevent 2.1.12-stable, OpenSSL 3.5.5, Zlib 1.3.1, Liblzma 5.8.2, Libzstd 1.5.7 and Glibc 2.42 as libc.  
Tor compiled with GCC version 15.2.0
```

- STEP 2 - Start Tor Service



```
kali@kali: ~  
$ sudo systemctl start tor@default  
$ sudo systemctl status tor@default  
● tor@default.service - Anonymizing overlay network for TCP  
   Loaded: loaded (/usr/lib/systemd/system/tor@default.service; enabled-runtime; preset: disabled)  
   Active: active (running) since Mon 2026-04-13 11:47:53 IST; 1min 10s ago  
  Invocation: 6068804a32324d8db60792b261706695  
    Process: 5584 ExecStartPre=/usr/bin/install -Z -m 02755 -o debian-tor -g debian-tor -d /run/tor (code=0)  
    Process: 5587 ExecStartPre=/usr/bin/tor --defaults-torrc /usr/share/tor/tor-service-defaults-torrc (code=0)  
   Main PID: 5589 (tor)  
      Tasks: 5 (limit: 6868)  
    Memory: 139.2M (peak: 139.4M)  
       CPU: 1.998s  
    CGroup: /system.slice/system-tor.slice/tor@default.service  
            └─5589 /usr/bin/tor --defaults-torrc /usr/share/tor/tor-service-defaults-torrc -f /etc/tor/torrc  
  
Apr 13 11:48:00 kali Tor[5589]: Bootstrapped 50% (loading_descriptors): Loading relay descriptors  
Apr 13 11:48:00 kali Tor[5589]: I learned some more directory information, but not enough to build a circuit  
Apr 13 11:48:06 kali Tor[5589]: The current consensus contains exit nodes. Tor can build exit and intermediate circuits  
Apr 13 11:48:20 kali Tor[5589]: Bootstrapped 55% (loading_descriptors): Loading relay descriptors  
Apr 13 11:48:21 kali Tor[5589]: Bootstrapped 60% (loading_descriptors): Loading relay descriptors  
Apr 13 11:48:21 kali Tor[5589]: Bootstrapped 67% (loading_descriptors): Loading relay descriptors  
Apr 13 11:48:21 kali Tor[5589]: Bootstrapped 75% (enough_dirinfo): Loaded enough directory info to build circuits  
Apr 13 11:48:21 kali Tor[5589]: Bootstrapped 90% (ap_handshake_done): Handshake finished with a relay to build a circuit  
Apr 13 11:48:21 kali Tor[5589]: Bootstrapped 95% (circuit_create): Establishing a Tor circuit  
Apr 13 11:48:22 kali Tor[5589]: Bootstrapped 100% (done): Done  
lines 1-23/23 (END)... skipping...  
● tor@default.service - Anonymizing overlay network for TCP
```

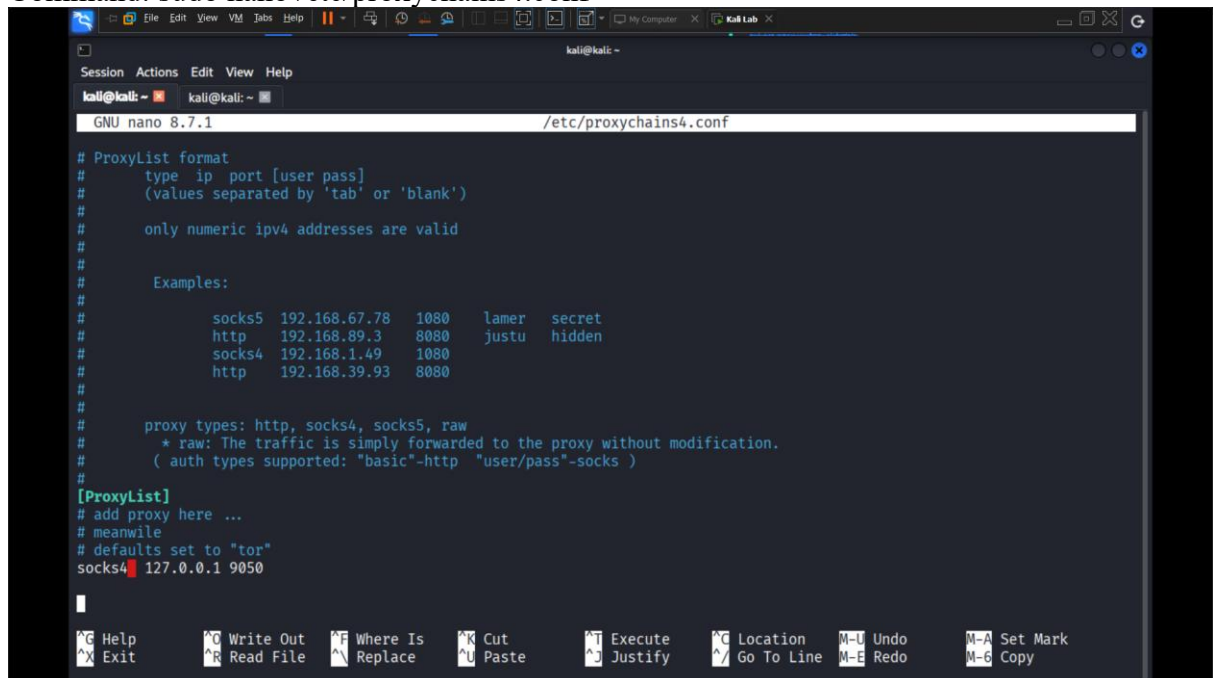
- STEP 3 - Configure ProxyChains
Check your normal IP



```
kali@kali: ~  
$ curl https://api.ipify.org  
103.10.28.204  
$
```

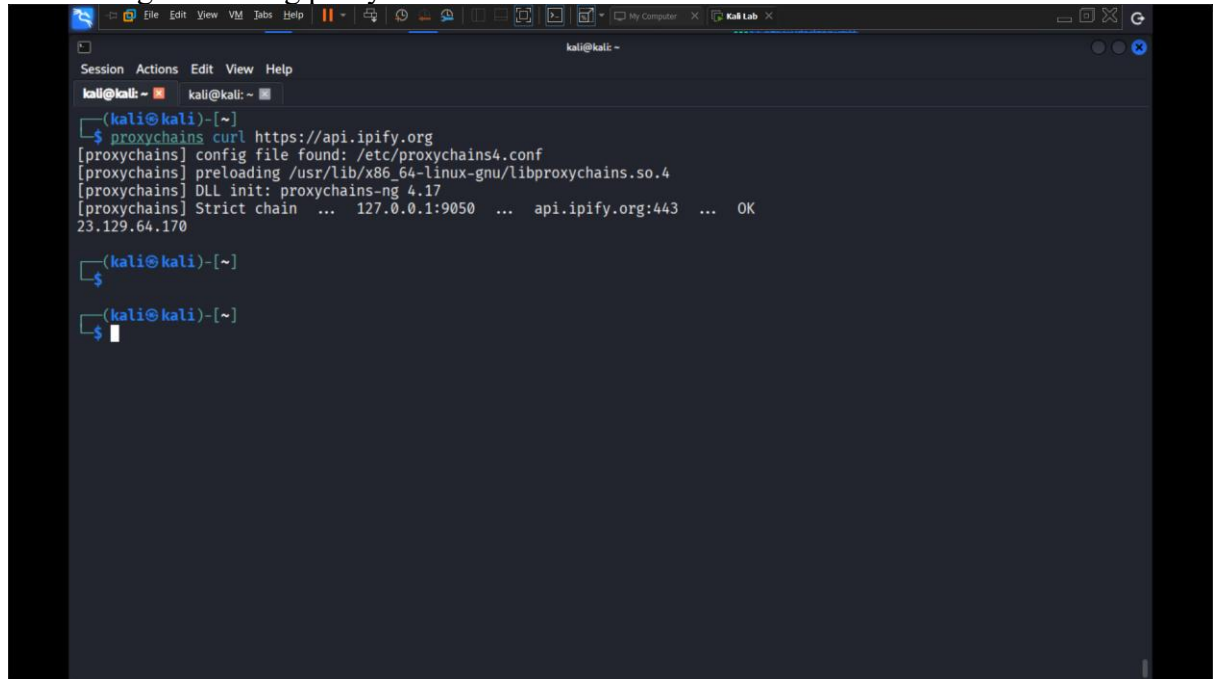
Configure ProxyChains

Command: `sudo nano /etc/proxychains4.conf`



```
GNU nano 8.7.1 /etc/proxychains4.conf  
  
# ProxyList format  
#   type ip port [user pass]  
#   (values separated by 'tab' or 'blank')  
#  
#   only numeric ipv4 addresses are valid  
#  
#   Examples:  
#  
#       socks5 192.168.67.78 1080 lamer secret  
#       http 192.168.89.3 8080 justu hidden  
#       socks4 192.168.1.49 1080  
#       http 192.168.39.93 8080  
#  
#   proxy types: http, socks4, socks5, raw  
#   * raw: The traffic is simply forwarded to the proxy without modification.  
#   ( auth types supported: "basic"-http "user/pass"-socks )  
#  
[ProxyList]  
# add proxy here ...  
# meanwhile  
# defaults set to "tor"  
socks4 127.0.0.1 9050  
$
```

Run through Tor using proxychains:



```
kali@kali: ~  
$ proxychains curl https://api.ipify.org  
[proxychains] config file found: /etc/proxychains4.conf  
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4  
[proxychains] DLL init: proxychains-ng 4.17  
[proxychains] Strict chain ... 127.0.0.1:9050 ... api.ipify.org:443 ... OK  
23.129.64.170  
  
kali@kali: ~  
$  
  
kali@kali: ~  
$
```

c. Task C3 - OPSEC Case Study
Case: Alexandre Cazes (AlphaBay Market)

Section	Answer
Who was the person / group?	Alexandre Cazes, the alleged operator of the darknet marketplace AlphaBay
What was the OPSEC mistake they made?	He reused a personal email address and username (“Alpha02”) linked to early AlphaBay account creation. He also left financial traces through cryptocurrency wallets and failed to fully anonymize hosting and infrastructure setup.
How were they identified / caught?	Investigators traced early server registration emails and cryptocurrency transaction links back to his real identity. Law enforcement coordinated international tracking and located him in Thailand, where he was arrested. He was later found dead in custody.
What should they have done instead?	He should have avoided any reuse of identifiers tied to real identity, used proper operational separation for emails and accounts, and ensured stronger anonymity for financial transactions and server infrastructure.
What OPSEC step did they fail? (1–5)	3. Financial and identity linkage failure (crypto + email correlation)